MSG No. 105.217 P_C4_2mc [Type IV, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `2mm`.

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,4,5,6]`
2	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[2,3,7,8]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[9,12,13,14]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[10,11,15,16]`

Table 2: Wyckoff site: `4b`, site symmetry: `2mm`.

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,4,5,6]`
2	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[2,3,7,8]`
3	$\left[\frac{1}{2},\ 0,\ z\right]$	`[9,12,13,14]`
4	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[10,11,15,16]`

Table 3: Wyckoff site: `8c`, site symmetry: `2'..`

No.	position	mapping
1	$\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$	`[1,12]`
2	$\left[\frac{3}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[2,11]`
3	$\left[\frac{1}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[3,10]`
4	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	`[4,9]`
5	$\left[\frac{1}{4},\ \frac{1}{4},\ z\right]$	`[5,14]`
6	$\left[\frac{3}{4},\ \frac{3}{4},\ z\right]$	`[6,13]`
7	$\left[\frac{3}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[7,16]`
8	$\left[\frac{1}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[8,15]`

Table 4: Wyckoff site: `8d`, site symmetry: `.m.`

No.	position	mapping
1	$[0,\ y,\ z]$	`[1,5]`
2	$\left[-y,\ 0,\ z+\frac{1}{2}\right]$	`[2,7]`
3	$\left[y,\ 0,\ z+\frac{1}{2}\right]$	`[3,8]`
4	$[0,\ -y,\ z]$	`[4,6]`
5	$\left[\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	`[9,13]`
6	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[10,15]`
7	$\left[y+\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[11,16]`
8	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ z\right]$	`[12,14]`

Table 5: Wyckoff site: `16e`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$\left[-y,\, x,\, z+\frac{1}{2}\right]$	`[2]`
3	$\left[y,\, -x,\, z+\frac{1}{2}\right]$	`[3]`
4	$[-x,\, -y,\, z]$	`[4]`
5	$[-x,\, y,\, z]$	`[5]`
6	$[x,\, -y,\, z]$	`[6]`
7	$\left[-y,\, -x,\, z+\frac{1}{2}\right]$	`[7]`
8	$\left[y,\, x,\, z+\frac{1}{2}\right]$	`[8]`
9	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z\right]$	`[9]`
10	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	`[10]`
11	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	`[11]`
12	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z\right]$	`[12]`
13	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z\right]$	`[13]`
14	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z\right]$	`[14]`
15	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	`[15]`
16	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	`[16]`